

Grade 11 Assessment 1 – B2.1 Programming Fundamentals

Mark Scheme

Total Marks: 40

Section A: Variables & Data Types

(8 marks)

Question 1

[4]

Value	Data Type
25	int
3.14	float
"Hello"	str
True	bool

Question 2

(a) Local (1). [1]

(b) A local variable can only be accessed within the function where it is declared (1). It does not exist outside that function's scope (1). [2]

Question 3

global (1). [1]

Section B: String Manipulation

(10 marks)

Question 4

[3]

Expression	Output
<code>word[0:3]</code>	<code>"Com"</code>
<code>word[3:]</code>	<code>"puter"</code>
<code>word[-3:]</code>	<code>"ter"</code>

Question 5

`Hello, Bill!` (2). **[2]**

Question 6

[2]

Line 1: `hello, world!`

Line 2: `Hello, IB!`

Question 7

[3]

Strings are immutable because they cannot be changed in place after they are created (1). Methods like `.replace()` and `.lower()` return a **new** string rather than modifying the original (1). Example: `message.replace("World", "IB")` returns a new string but `message` remains unchanged (1).

Section C: Exception Handling

(10 marks)

Question 8

[3]

Keyword	Purpose
<code>try</code>	Contains code that might raise an exception (1)
<code>except</code>	Catches and handles specific types of errors (1)
<code>finally</code>	Executes code no matter what, whether an exception was raised or not (1)

Question 9

(a) Please enter a valid integer. (1). [1]

(b) You cannot divide by zero. (1). [1]

(c) `except Exception` catches all exceptions (1). If placed first, it would catch everything and the more specific handler would never run (1). [2]

Question 10

[3]

Place the file-reading code inside a `try` block (1). Use `except FileNotFoundError` to catch the error if the file does not exist (1). Display a helpful message to the user instead of crashing (1).

Section D: Debugging Techniques

(12 marks)

Question 11

[4]

(a) Syntax error: Violation of the rules of the programming language (1). Example: missing parentheses or colon (1).

(b) Logical error: Code runs without crashing but produces incorrect results (1). Example: using wrong formula or incorrect loop range (1).

Question 12

(a) [3]

Iteration	i	total
1	0	0
2	1	1
3	2	3

(b) 3 (1). [1]

Question 13

[2]

Insert `print(price)` inside the loop to see each value (1). This would reveal that `"30"` is a string, not an integer (1).

Question 14

[2]

A breakpoint is a marker that pauses execution at a specific line (1). It allows the programmer to inspect variable values and step through code line by line to find the error (1).

Grade Boundaries (Suggested)

Grade	Mark Range
A*	36–40
A	32–35
B	28–31
C	24–27
D	20–23
E	16–19
U	Below 16

Alignment with IB Computer Science Guide (First Assessment 2027)

Syllabus Point	Description	Questions
B2.1.1	Construct and trace programs using global/local variables and data types	1, 2, 3, 12
B2.1.2	Construct programs that extract and manipulate substrings	4, 5, 6, 7

Syllabus Point	Description	Questions
B2.1.3	Describe how programs use common exception handling techniques	8, 9, 10
B2.1.4	Construct and use common debugging techniques	11, 12, 13, 14

Assessment Objectives addressed

- **AO1** (Knowledge & Understanding): Questions 1, 2, 3, 7, 8, 11, 14
- **AO2** (Apply & Use): Questions 4, 5, 6, 9, 10, 12, 13
- **AO3** (Construct, Analyse, Evaluate): Questions 9(c), 10, 12, 13

Command terms used: State, Describe, Explain, Distinguish, Complete, Identify